

What's next in Medical AI?

医学人工智能，路在何方？

From the early days of Pattern Recognition to the Future of AI

自模式识别之发轫，至人工智能之未来

Andreas Maier

About the Speaker 主讲人简介



Prof. Dr.-Ing. Andreas Maier

Responsibilities & Experience

- Head of the Pattern Recognition Lab, FAU
- Head of the Competence Center Medical Engineering Erlangen
- Head of the Medical AI Hub of Bavaria
- Member, AI Council of the Free State of Bavaria
- Scientific Counselor, Fraunhofer Institute
- AI Council, German Radiation Protection Society
- European Research Council (grantee & reviewer)
- Visiting Professor at Stanford University and MIT

Industry & Entrepreneurship

- Siemens Healthineers (innovation manager / tech lead)
- Start-ups: *Mira Vision* (shareholder, valued at EUR 10 M); *Norizon* (council, early VC phase)
- Holds more than 15 active US and EU patents

Theory first, application always.

- Founded in 1975 by Heinrich Niemann.
- Germany's first dedicated Pattern Recognition / Machine Learning lab.
- Medical imaging research as early as 1981.
- First computer: a **PDP-11/34** – 248 kB, 2.5 MHz.
- Focus from day one: sound theory and a deep understanding of the hardware.



PDP-11/34 – the lab's first computer.

Our Team Today 群策群力



The Pattern Recognition Lab 模式识别实验室

People

- Head: Prof. Dr.-Ing. Andreas Maier
- 2 emeritus professors
- 60+ PhD students and 8 postdocs
- 3 000 master's and bachelor's students each semester
- Secretaries (Alexandra H., Claudia S.) and our technician (Sven G.)

Partners

- Academic partners across Europe, Asia, the Americas and Australia
- Industry partners: Siemens, Fraunhofer, Adidas, Intel, Audi, IBM, and more

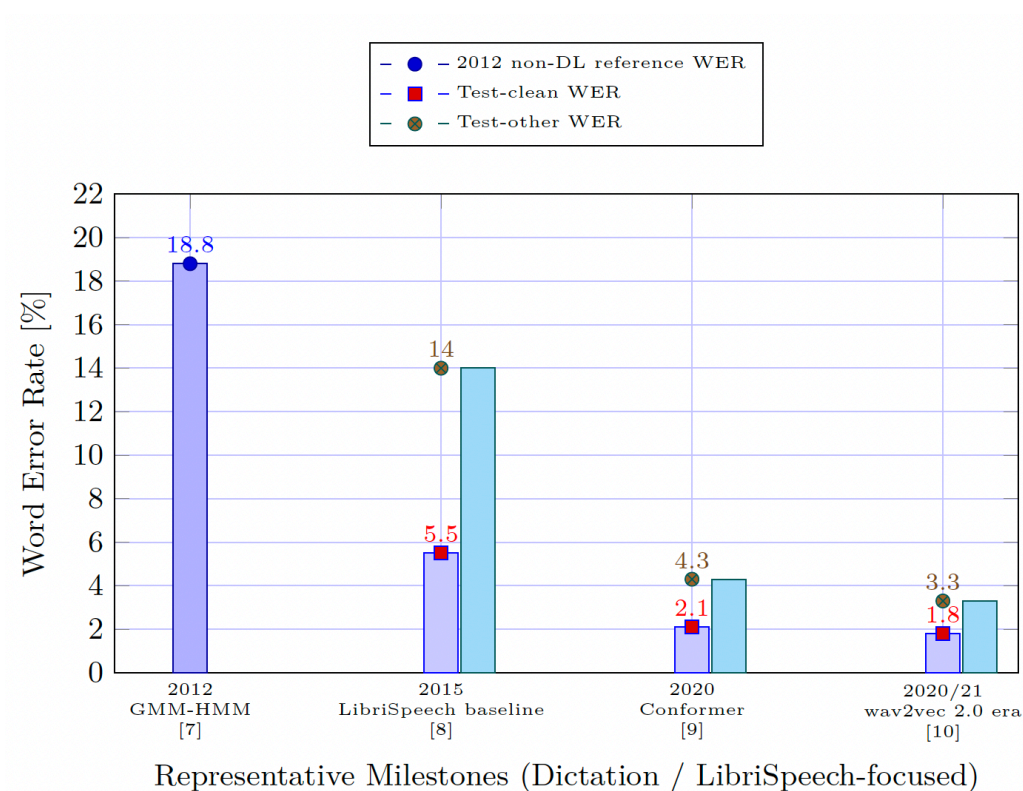
One lab, many hands, one shared effort.

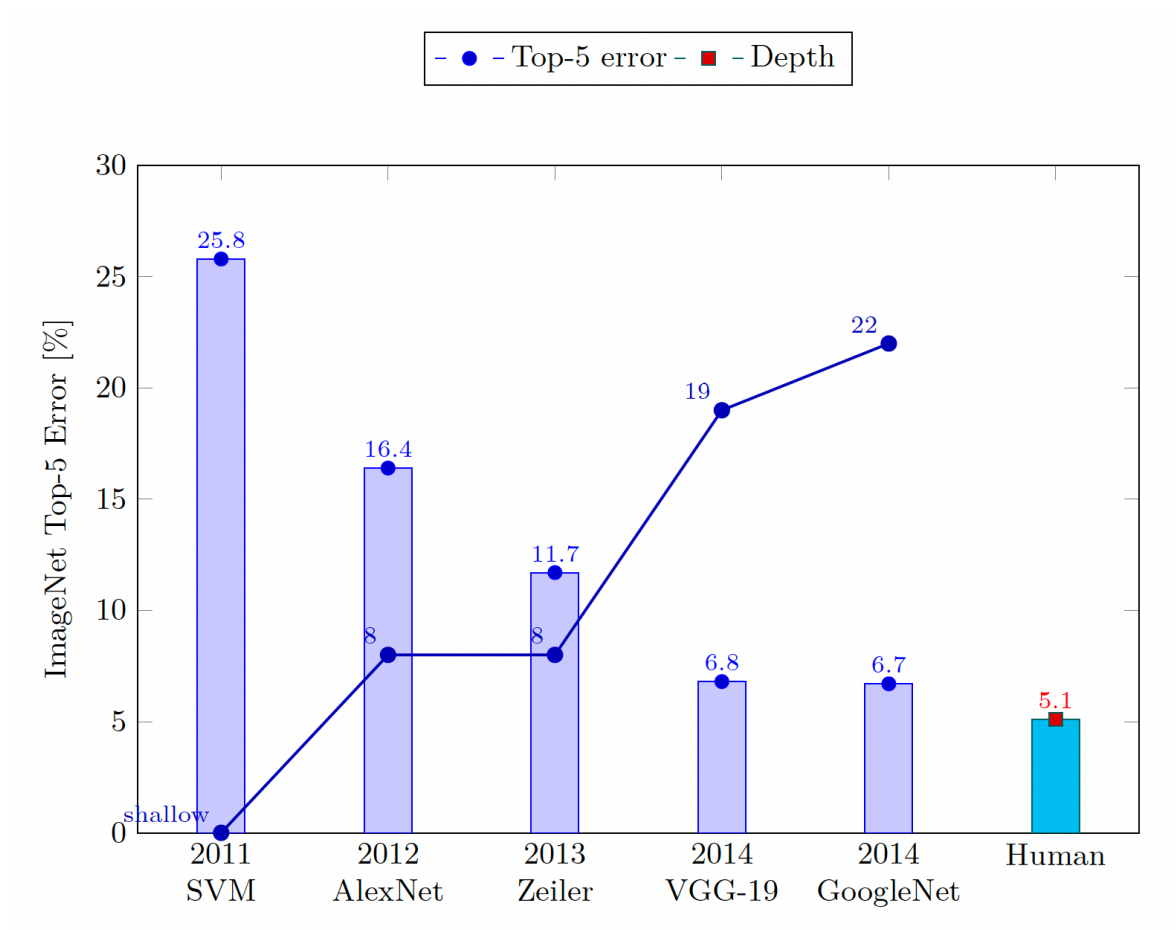
-
- What is deep learning?
 - What is agentic AI?
 - What is coming next?

What is deep learning?

何谓深度学习?

- Revolutionising pattern recognition and AI for more than 10 years.





Deep Learning 深度学习

Learning to Play 无师自通



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.

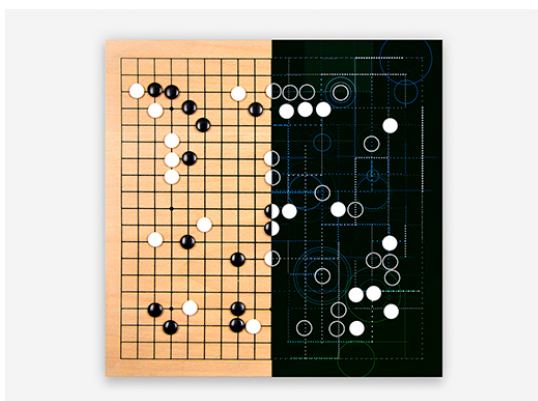
Deep Learning 深度学习

Learning to Play 无师自通



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.



AlphaGo (DeepMind, 2016)

- Self-play + Monte-Carlo tree search.
- Defeats world champion Lee Sedol 4–1.

Generative AI 生成式 AI

Source Image 原始图像





“An athletically built man with a serious expression stands in a well-lit room with a wooden ceiling. He is wearing a blue t-shirt with the phrase ‘SAVE THE CHUBBY UNICORNS’ and an image of a rhinoceros. The man is taking a selfie with a modern smartphone, capturing his reflection in a mirror. In the background, one can see gym equipment including dumbbells and a weight rack. There is also a hanging chandelier with a unique design, illuminating the room.”

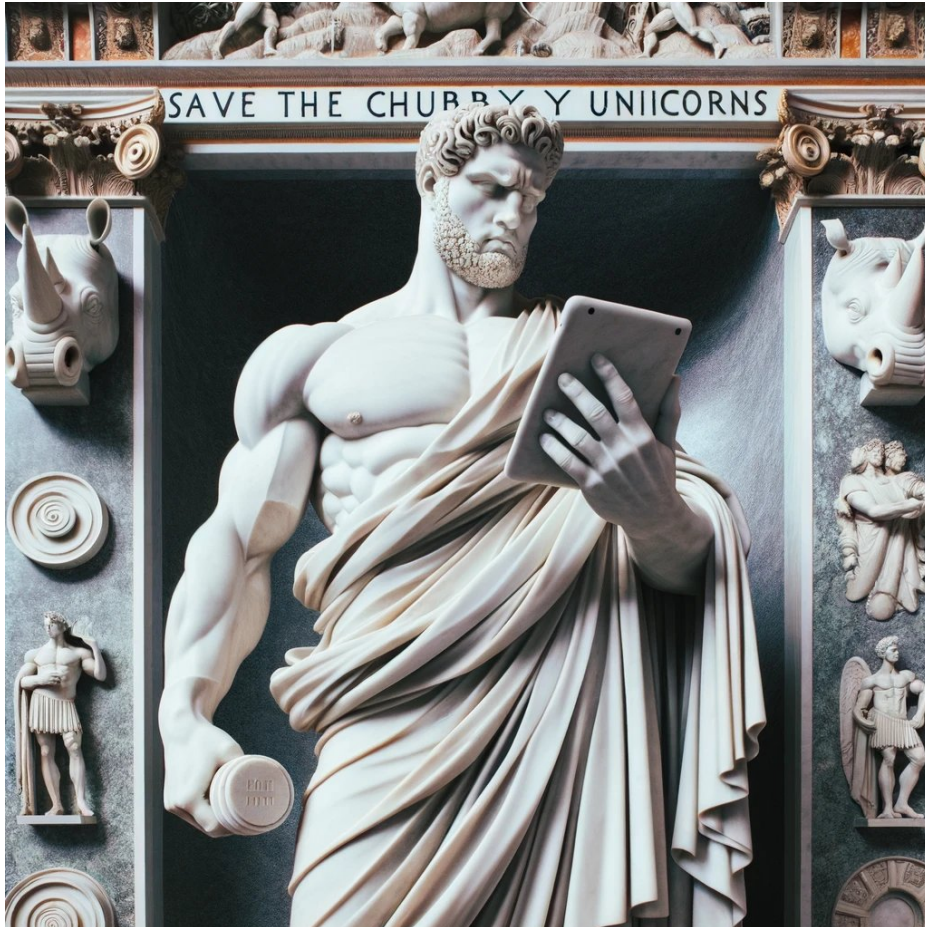
Generative AI 生成式 AI

Generated Image 生成图像



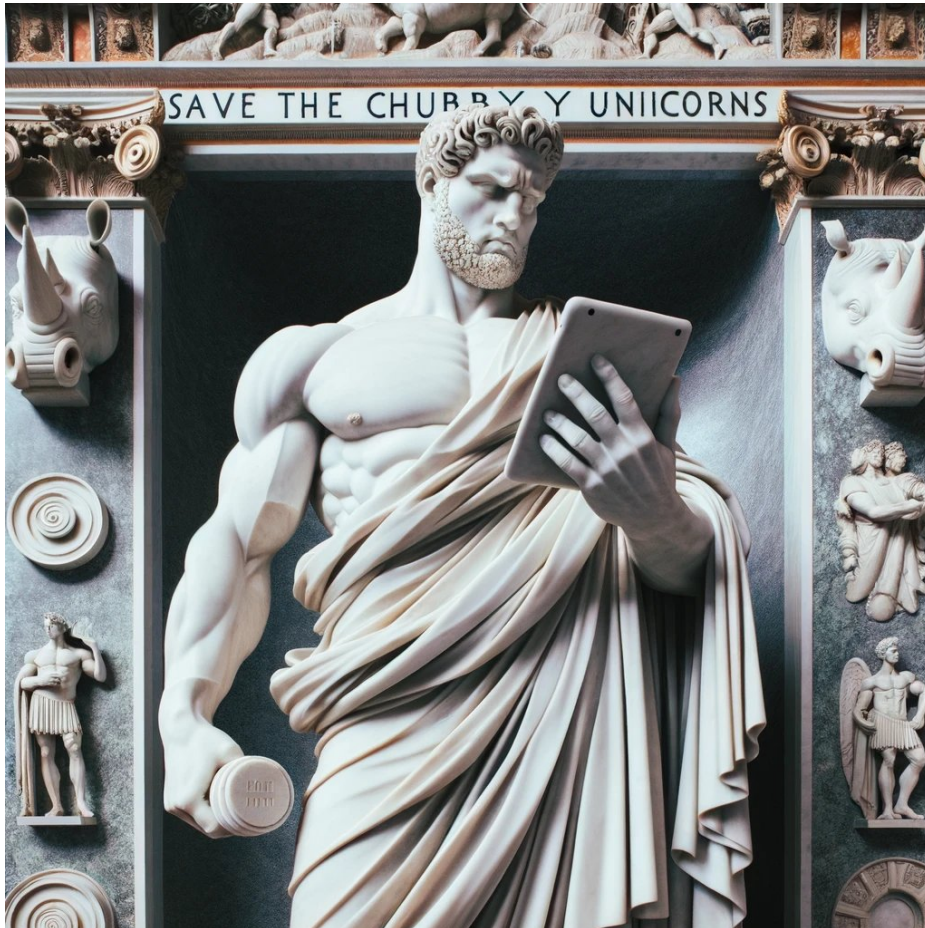
Generative AI 生成式 AI

Style Variations 风格变换



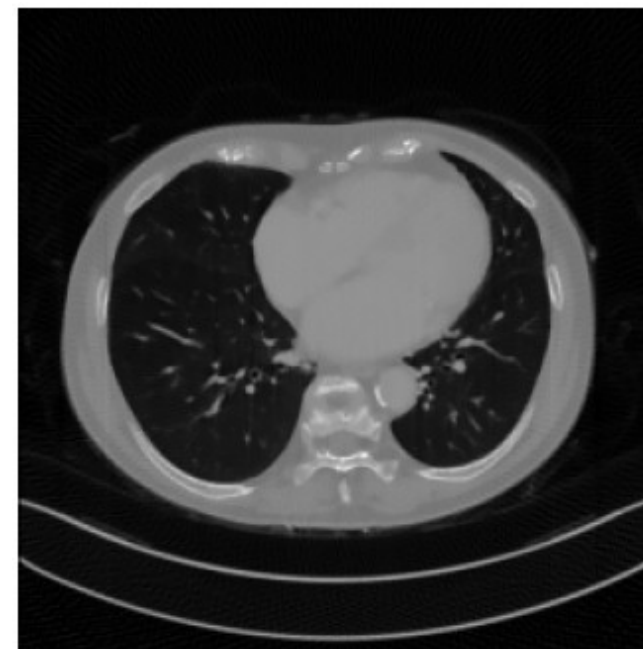
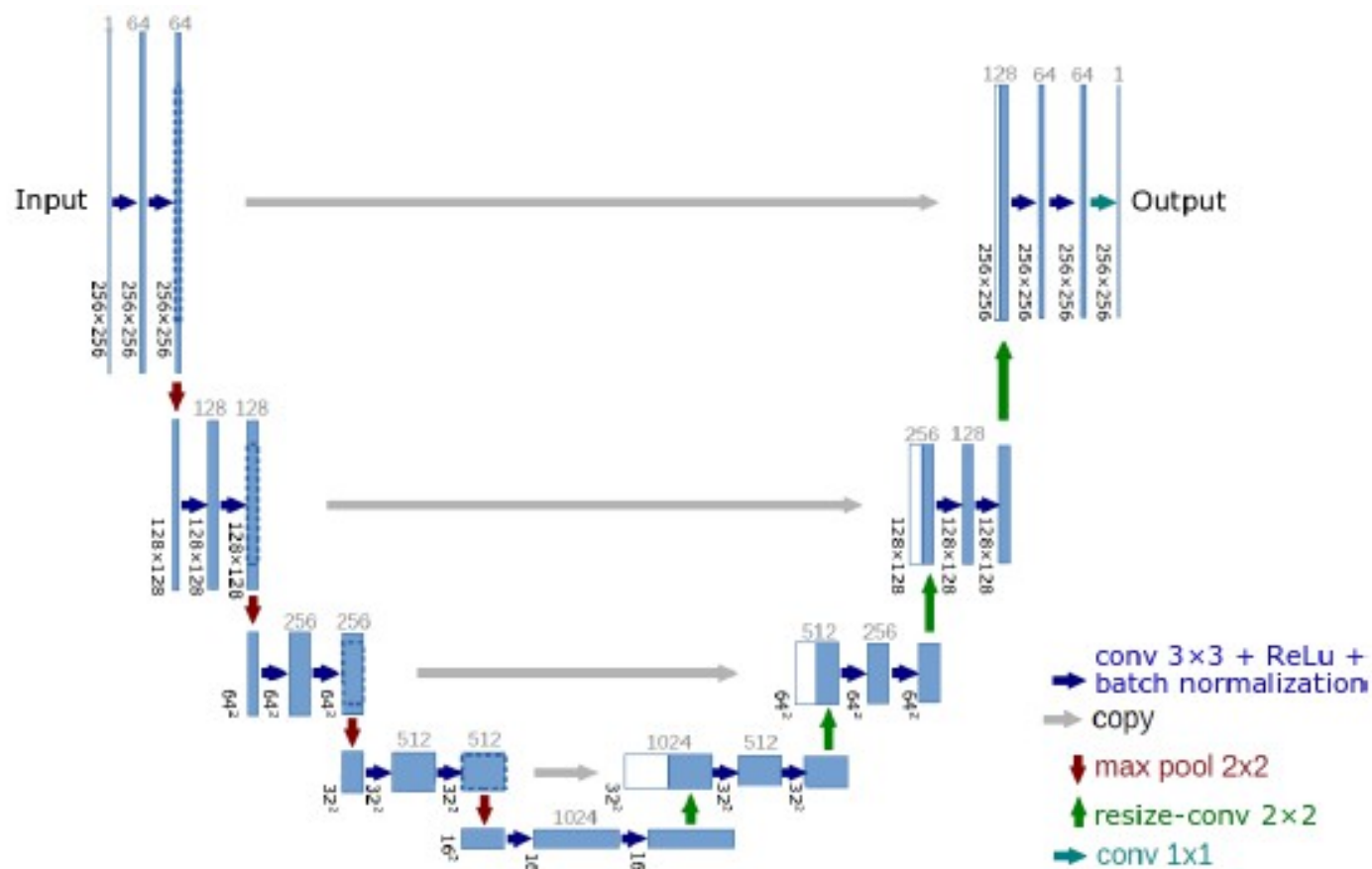
Generative AI 生成式 AI

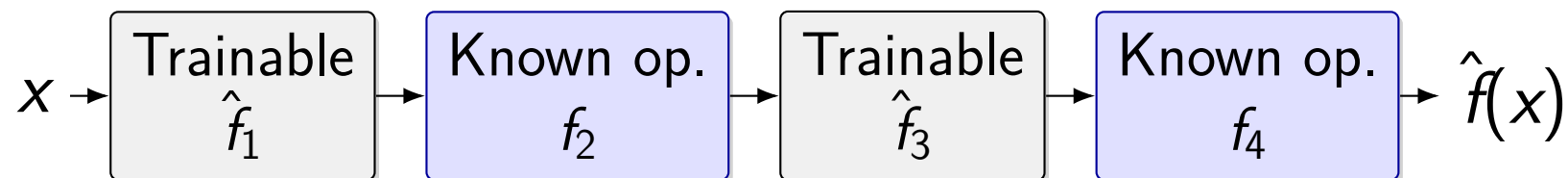
Style Variations 风格变换



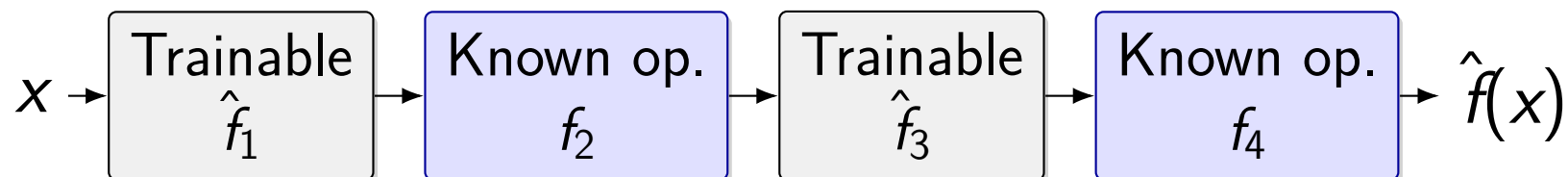
Medical Deep Learning 医学深度学习

From Segmentation to Reconstruction 从分割到重建





Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).



Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

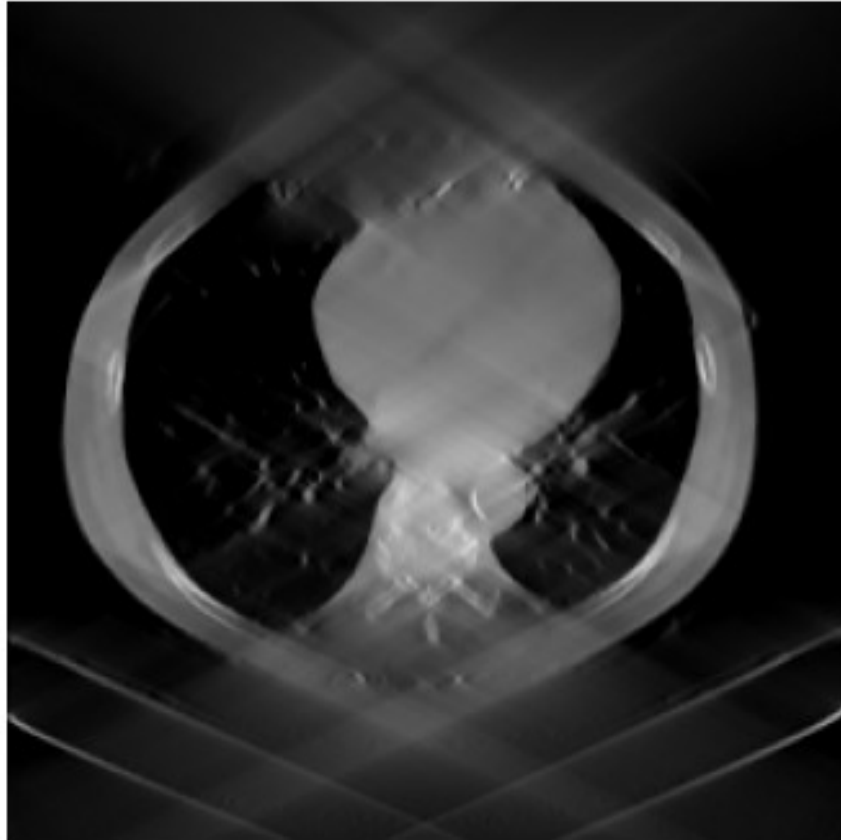
Theorem – known operators reduce the error bound

For $f = f_n \circ \dots \circ f_1$ approximated by $\hat{f} = \hat{f}_n \circ \dots \circ \hat{f}_1$ with per-layer errors ε_i and Lipschitz constants $\hat{L}_j$,

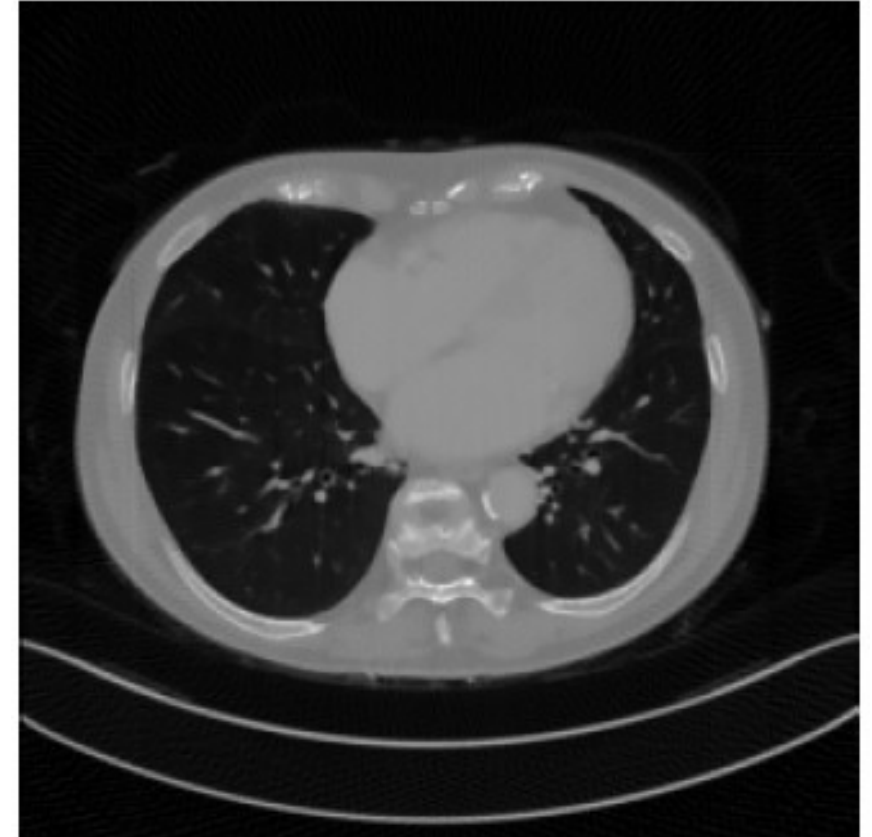
$$\varepsilon(\hat{f}) \leq \sum_{i=1}^n \left(\prod_{j=i+1}^n \hat{L}_j \right) \varepsilon_i.$$

Replacing $\hat{f}_k$ with the *known* f_k forces $\varepsilon_k = 0$ and strictly tightens this bound.

Maier et al., *Nature Machine Intelligence*, 2019.



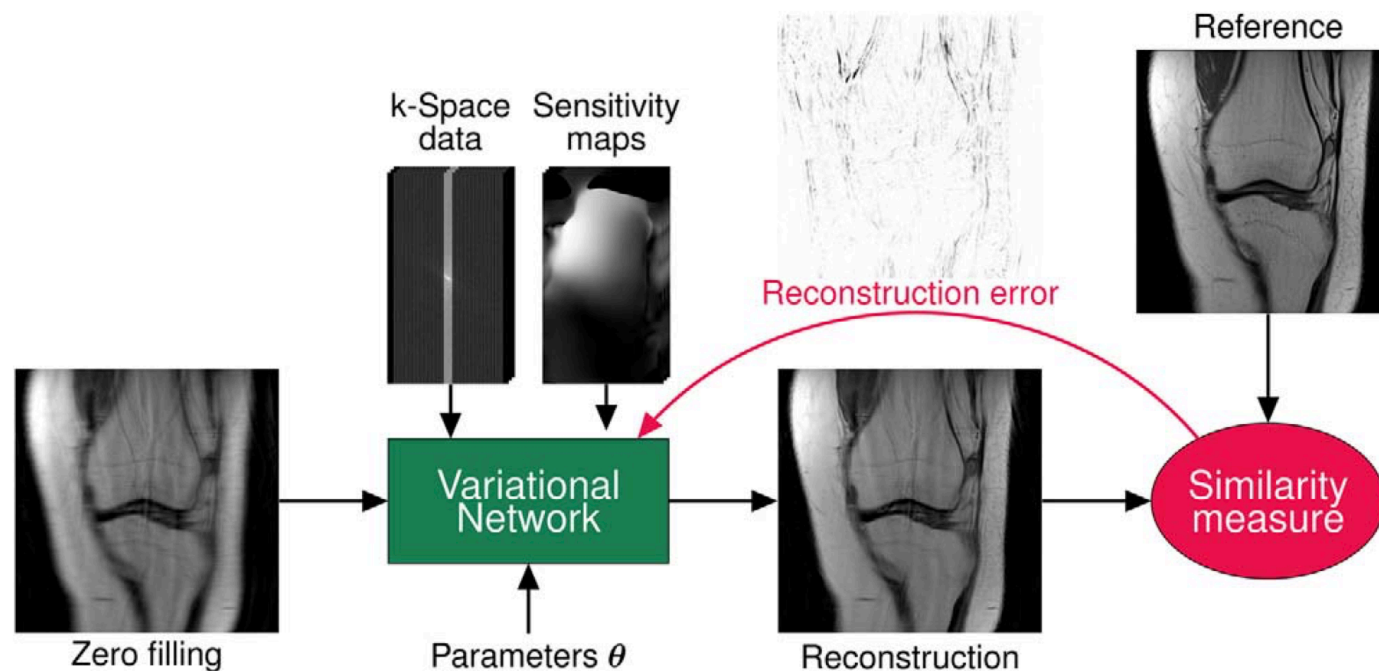
Conventional



With deep learning

Learning a Variational Network for Reconstruction of Accelerated MRI Data

Kerstin Hammernik,^{1*} Teresa Klatzer,¹ Erich Kobler,¹ Michael P. Recht,^{2,3}
Daniel K. Sodickson,^{2,3} Thomas Pock,^{1,4} and Florian Knoll^{2,3}



Medical Deep Learning 医学深度学习

Recognition 殊荣



Deutscher Zukunftspreis – low-field MRI made possible by deep learning.

What is agentic AI?

何谓智能体？

“How do you drive a car?”

Thought for a few seconds...

- A classic prompt to a large language model.
- After a short “thinking pause” the answer arrives as long-form running text.
- No tools, no actions – the model just *describes* how to drive.

To drive a car, follow these steps:

- **Get in and adjust:** fasten your **seat belt**, set the **seat** and **mirrors** so you can see the road clearly.
- **Start the engine:** press the **brake pedal**, turn the key or push the start button. Make sure the gear selector is in **Park (P)** or **Neutral (N)**.
- **Engage a gear:** for an automatic, move the lever to **Drive (D)**; for a manual, press the **clutch** and select **1st gear**.
- **Release the parking brake** and slowly lift your foot off the brake. Gently press the **accelerator** to move forward.
- **Steer** smoothly with both hands and **check mirrors** regularly. Keep a safe distance from the vehicle in front.
- **To stop:** ease off the accelerator, press the brake, and bring the car to a complete stop. Shift to **Park** and apply the parking brake before turning off the engine.

Always observe traffic rules and adapt your driving to road and weather.

⇒ **Lots of words, but no *action*.**

Agentic AI 智能体

From Language Model to Agent 从语言模型到智能体



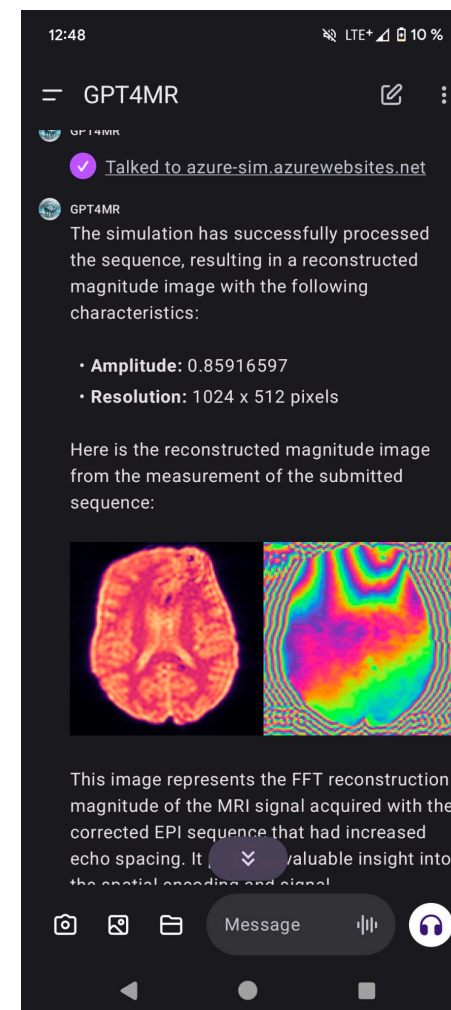
Moritz Zaiss

Agentic AI 智能体

From Language Model to Agent 从语言模型到智能体



Moritz Zaiss



Zaiss et al., *GPT4MR*, 2024.

“Code a spin echo EPI (64×64 , $fov = (0.2, 0.2, 0.08) m$, $TE = 150 ms$)!”

- **Spin-echo EPI** – an advanced MRI sequence used for diffusion and functional imaging.
- Three models receive the *same* natural-language instruction and must produce runnable Python code.
- Contestants:

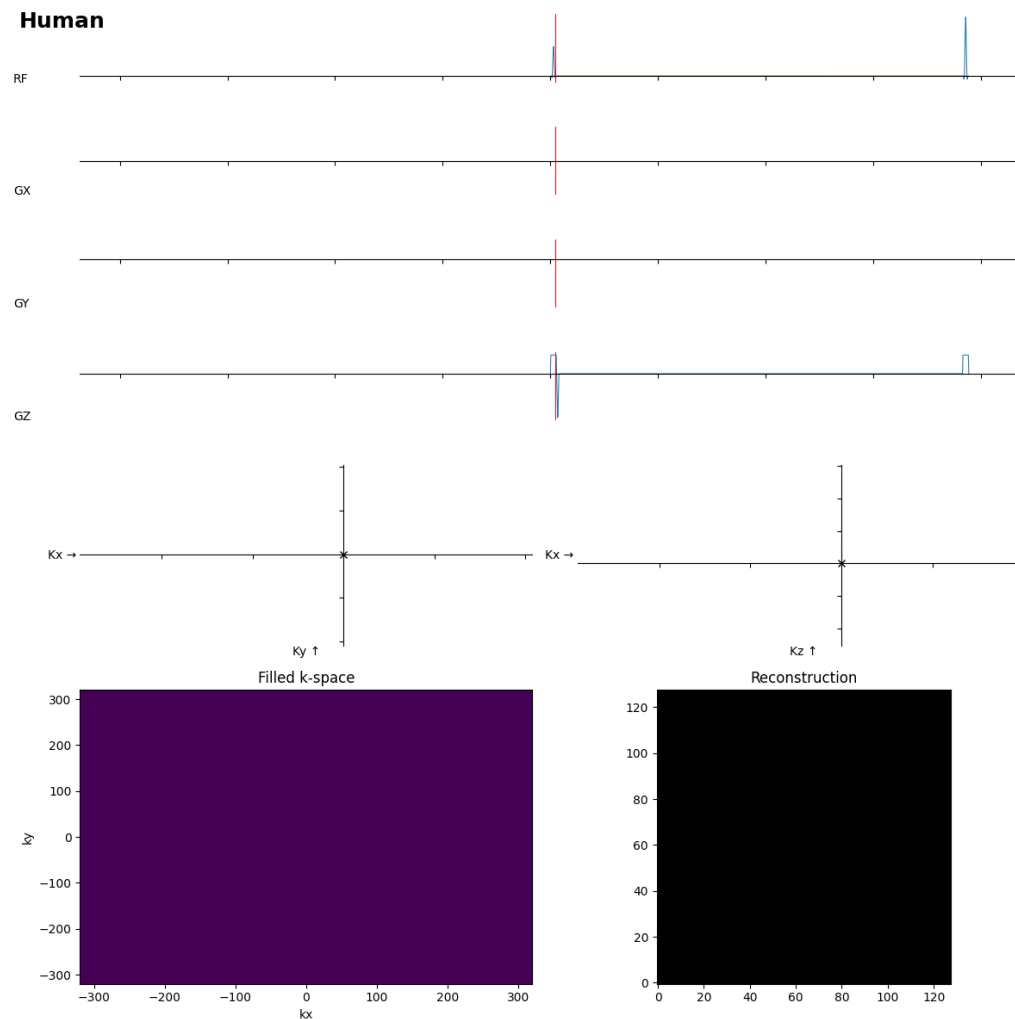
Human
reference

GPT-4o
OpenAI

**GROK-
3-think**
xAI

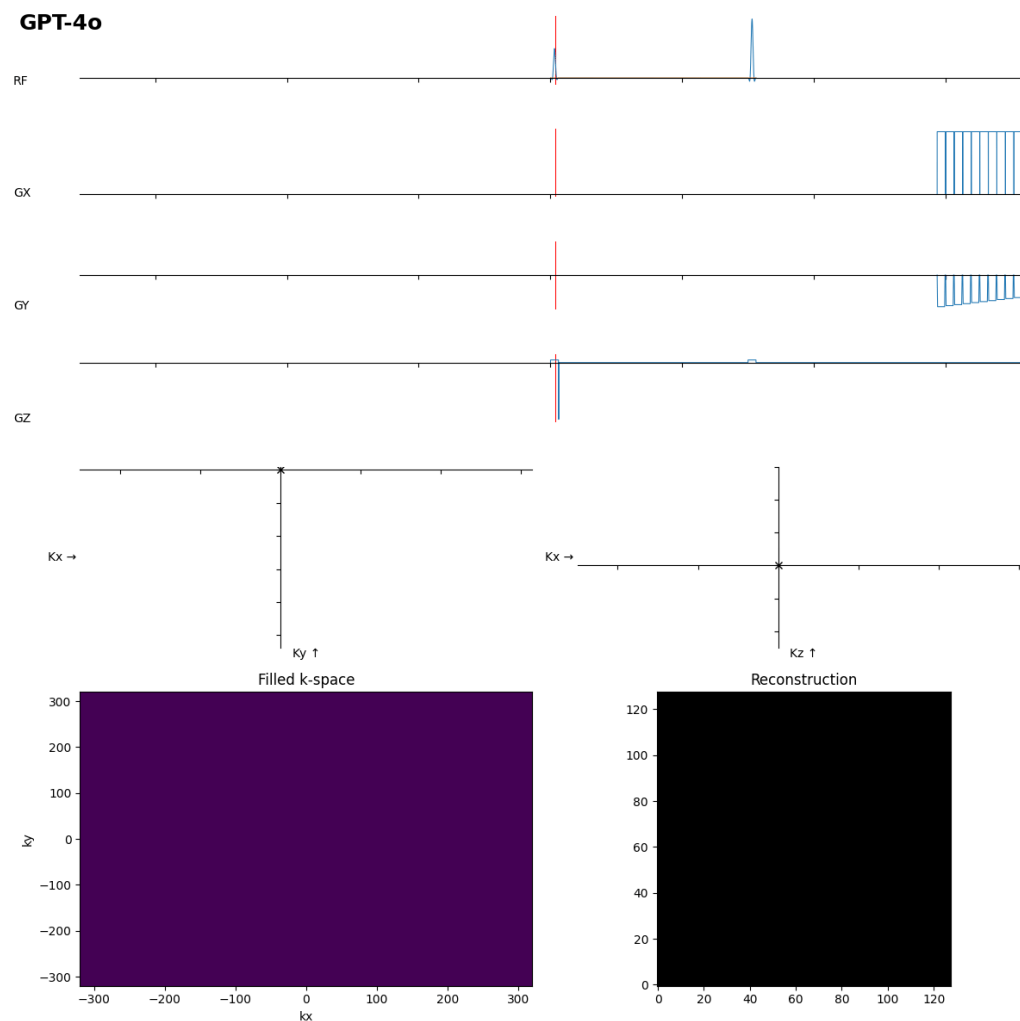
Agentic AI 智能体

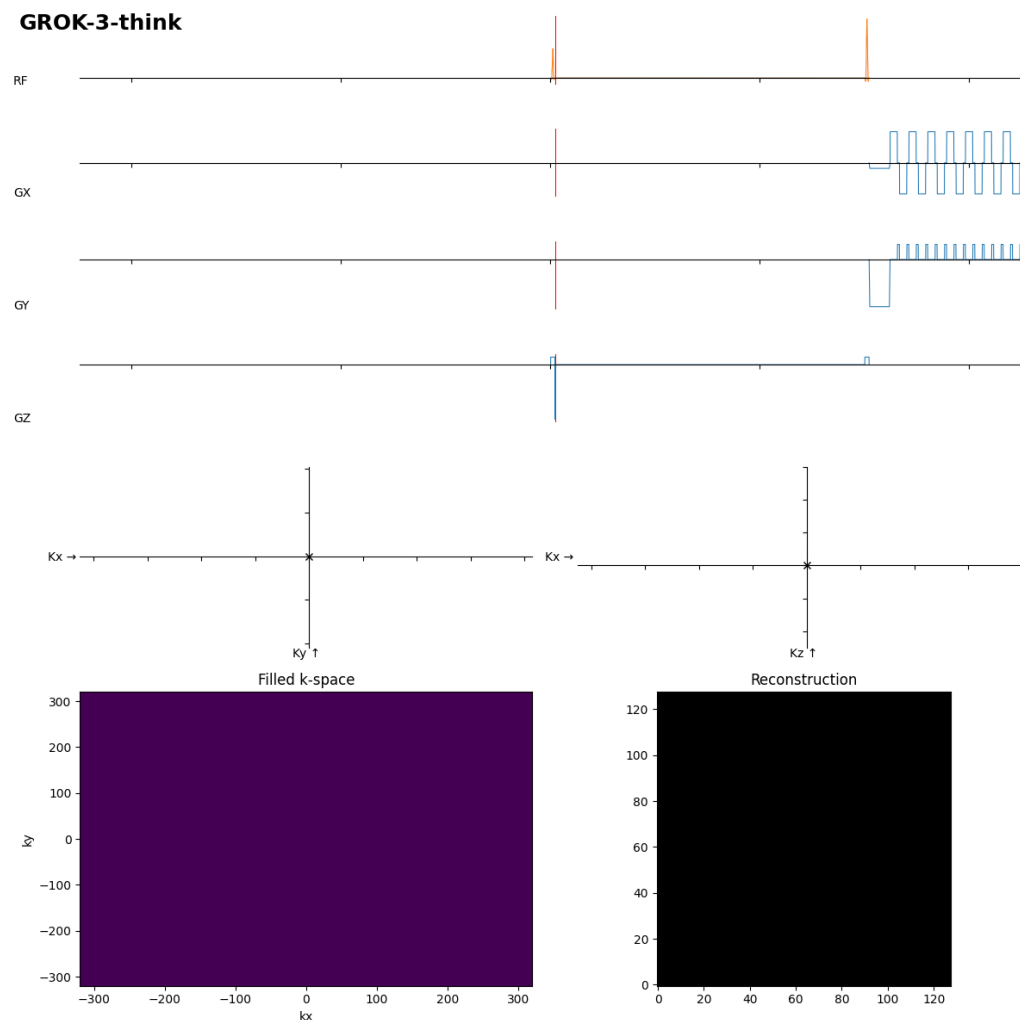
Human Baseline 人类基准



Agentic AI 智能体

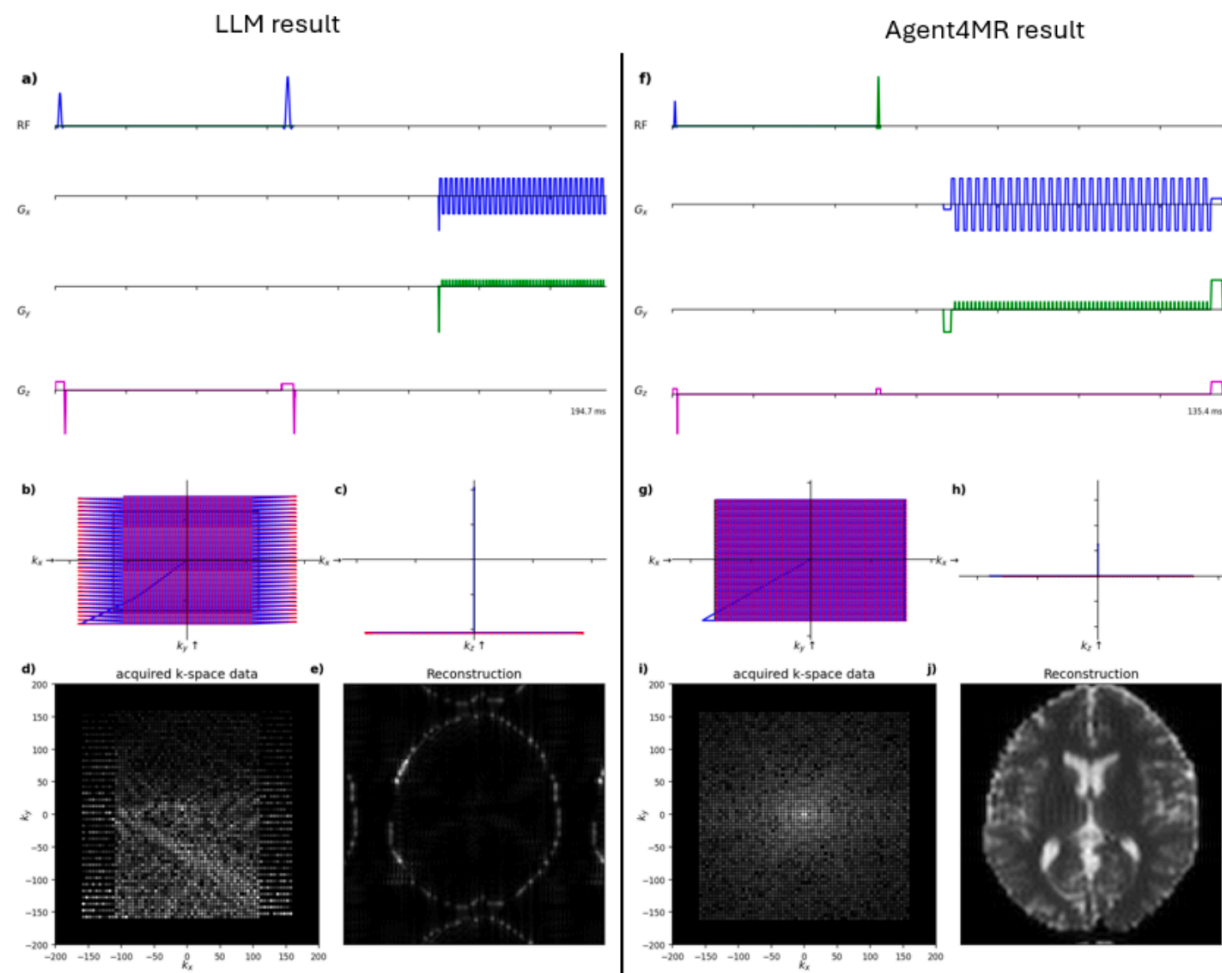
GPT-4o Attempt GPT-4o 的尝试





How about real agentic AI?

何为真正的智能体？



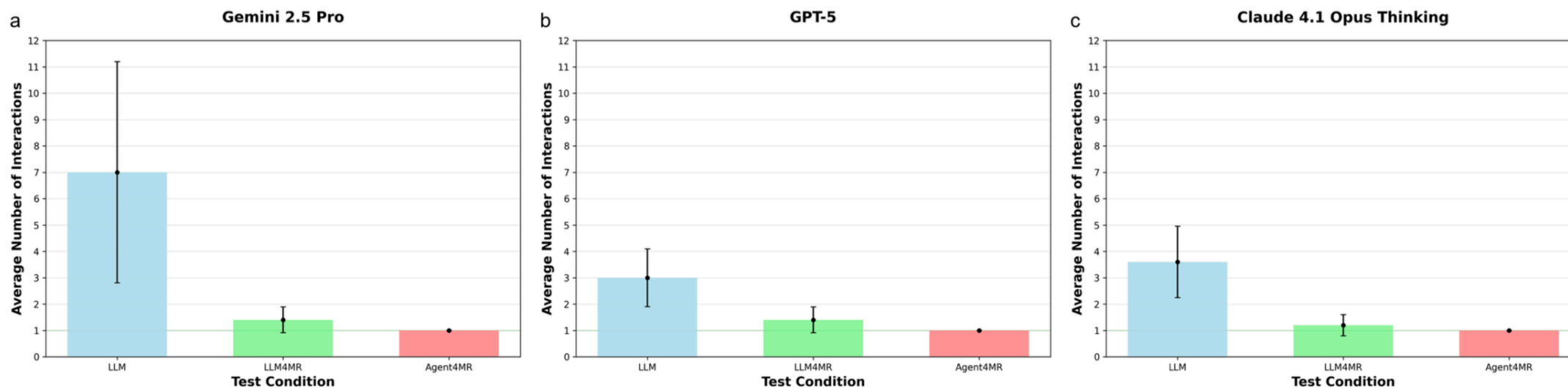


Table 1: User interactions N, time, token usage, and estimated cost until a satisfactory sequence. Human baseline: recently trained EPI-naive student, 2.5 h, 5 interactions.

Configuration	N	Time	Tokens	Cost [€]
Agent4MR		[min]		[≈]
Gemini 2.5 Pro	1	4.8	87k	0.12
GPT-5	1	15	520k	0.48
Claude 4.1 Opus	1	10.2	1162k	10.00
Human	5	150	—	37.50

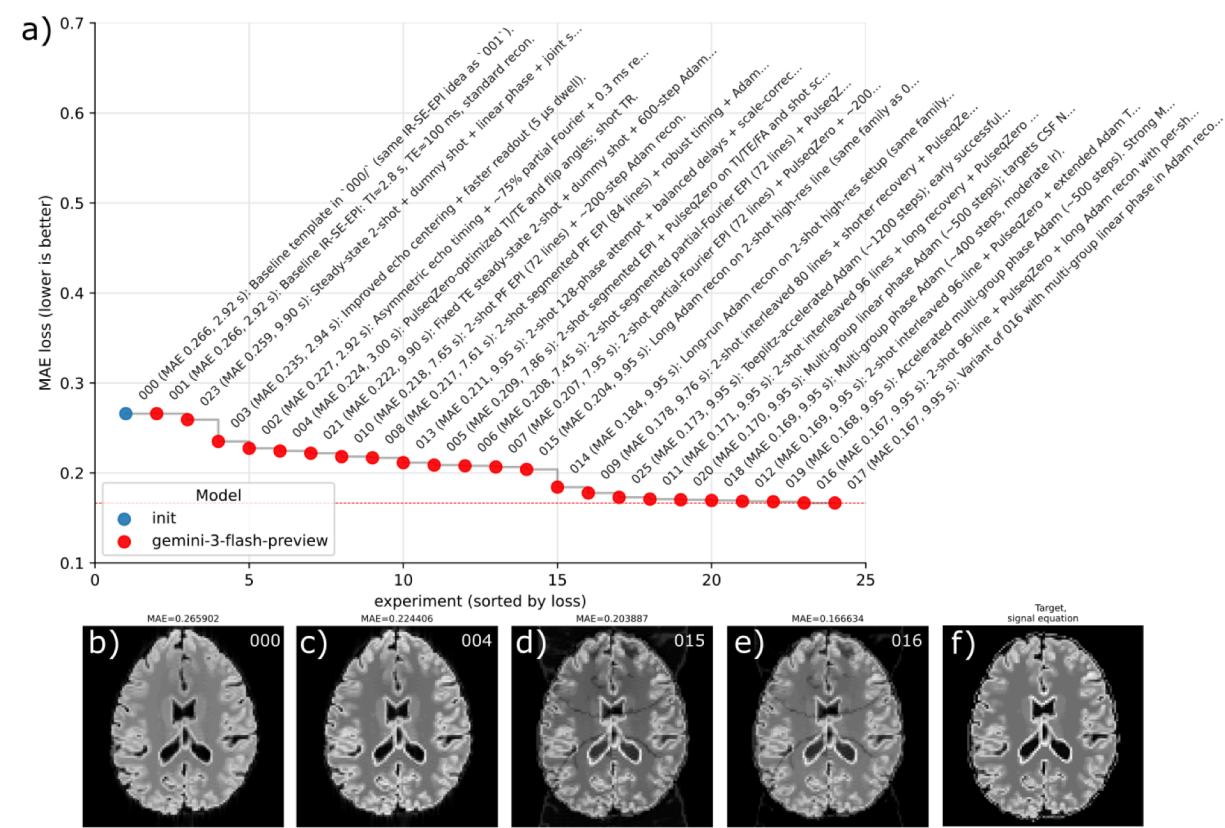


Figure 5: IR-SE-EPI MAE leaderboard progress. Autonomous agents iteratively improved the IR-SE-EPI pipeline (sequence parameters, reconstruction, post-processing). Experiments are ranked by MAE loss (worst to best). The staircase shows improvement from baseline (~0.2659) to best (~0.1666) through multi-window filtering, multi-shot introduction, phase correction, and differentiable TI/TE optimization.

Table 2: Normalized Token Usage and API Costs in EUR (n=25 events per model)

Gemini Model	Total Tokens [Mio]	Tokens per Exp. [Mio]	Cost [€]
2.5 Flash	25.456	1.018	2.43
3 Flash	136.340	5.453	12.85
3.1 Pro	72.039	2.881	59.72
Total	233.836	9.353	75.00

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www.ILMTALKLINIK.DE

Patientendaten/Aufkleber

Thieme Compliance ID 15Th GB

Coronary Angiography, Coronary Angioplasty and Stent Implantation
(Examination of Coronary Blood Vessels, Including Possible Dilatation of Constrictions/Implantation of a Stent)

Hello,

The purpose of this informed consent leaflet is to help you prepare for the patient-doctor discussion. Please read it carefully before the discussion and complete the questionnaire carefully and completely. For better readability, we use male pronouns for occupational titles or terms referring to specific persons (e.g. doctor) but are including all genders with them.

Which procedure is planned?

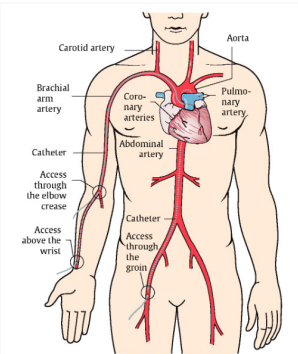


Fig. 1: Possible catheter access via the groin, above the wrist and in the crook of the arm

Current findings and your symptoms lead to suspicion of coronary artery disease.

This needs to be evaluated by a special X-ray examination (coronary angiography). If any constrictions (stenoses) or obstructions are found in the coronary vessels, it may be possible to perform an immediate dilatation or to reopen them using a balloon catheter (coronary angioplasty), and a stent (a tube to hold the vessel open) can be implanted if needed (stent implantation).

Are alternative examination/treatment methods available?

Other non-invasive imaging techniques to visualise the coronary blood vessels, e.g. computed tomography (CT) and magnetic resonance imaging (MRI), are possible but are less precise and do not enable the doctor to perform the necessary treatment measures during the same session.

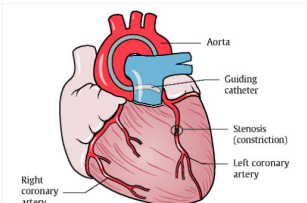


Fig. 2: Representation of coronary blood vessels showing the constriction

ID: 15THGB-11/2023v1-01/2024ex-Daten: 24.03.2025-Druck: 11.10.2025(11:38 Uhr) Seite 1/7

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FAU A. Maier What's next in Medical AI?

30 May 2026 ▪ West China Hospital ▪ 华西医院骨科八十周年 32/49

Agentic AI 智能体

Adaptive Systems 自适应系统

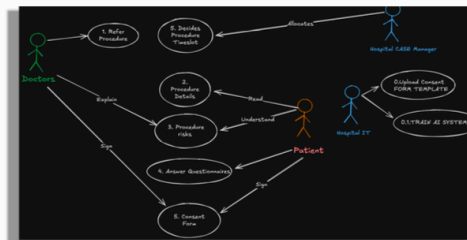


AI-POWERED & ADAPTIVE:
Intelligent, Personalize Interaction

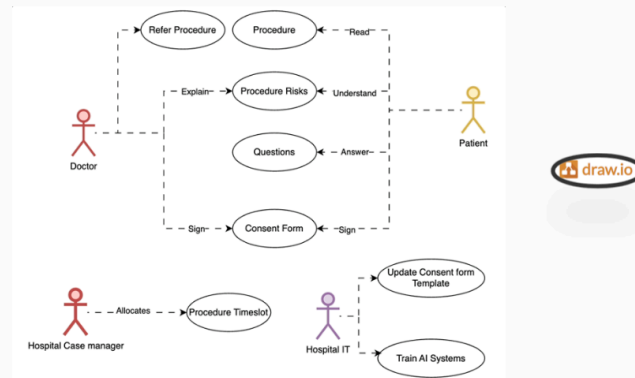
Geek Box 1: Draw.io: from sketch to formal diagram

Draw.io (also known as diagrams.net, <https://app.diagrams.net>) is a free, browser-based diagramming tool. Diagrams can be exported as XML, SVG, or PNG and version-controlled alongside code. Draw.io also supports AI-assisted diagramming: configurable large language model (LLM) backends—including OpenAI, Anthropic, and local models—can generate, modify, and explain diagrams from natural-language prompts (<https://www.drawio.com/doc/faq/configure-ai-options>).

The two figures below show the same actor model: first sketched by hand on an iPad, then redrawn in Draw.io. The content is identical—the difference is the tool.



(a) iPad sketch: unstructured brainstorming, needs vision-language model (VLM) for access.



(b) Draw.io: version-controllable, exportable, and AI-editable.

Agentic AI 智能体

From Sketch to Diagram 从草图到图表



The screenshot displays a Replit workspace for a project named "CardiacConsent". The interface is split into three main sections: a left sidebar with project management tools, a central preview window, and a right sidebar with a console and chat.

Left Sidebar:

- Project name: CardiacConsent
- Buttons: Upgrade, Add core components for informed consent web...
- Agent 3 chat window:
 - Message: "Link screen to real back end" (5/7)
 - Message: "Asked for OPENAI_API_KEY"
 - Text: "To enable the AI-powered Q&A feature that answers patient questions based on the consent document, I need an OpenAI API key. This key allows the application to use OpenAI's language models to provide intelligent, context-aware responses to patient questions about the coronary angiography procedure."
 - Text: "You can get an API key by:"
 - List:
 1. Creating an account at <https://platform.openai.com/>
 2. Going to API Keys section
 3. Creating a new secret key
 - Text: "Once you have the key, please provide it as OPENAI_API_KEY and I'll integrate the AI chat functionality."
 - Text: "1 secret requested • See all secrets"
 - Text: "OPENAI_API_KEY"
 - Text: "Would you like to save this Secret to your Account Secrets so you can easily access it in other Apps?"
 - Buttons: + Add to Account Secrets, Scroll to latest
 - Bottom button: Upgrade to continue building

Central Preview Window:

- Address bar: .replit.dev/
- Page title: Secure Patient Login
- Subtitle: Two-factor authentication for your safety
- Form fields:
 - Email Address: mario.sicaja@gmail.com
 - Send Verification Code button
- Footer: Demo Mode: Use any email and code 123456

Right Sidebar:

- Search, User, Publish, and Theme icons.

Friedrich-Alexander-Universität
Technische Fakultät



Vibe Coding: Software Engineering with LLMs

Introduction

A. Maier, S. Mei, A. Panambur, A. Sindel, S. Zeitler

Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg

April 13, 2026

Advanced AI-Assisted Techniques



Claude Skills [3]:

- Shareable workflow automation (YAML + Markdown)
- Team-wide best practices as code
- Deployment, testing, code review
- Available at github.com/anthropics/skills

Model Context Protocol (MCP):

- Standardized tool integration
- Connect AI to databases, APIs, files
- Growing ecosystem of plugins

Other Advanced Patterns:

- **Agentic workflows:** Multi-step autonomous tasks
- **Tool-augmented coding:** AI uses external tools
- **Vision support:** Screenshot-driven debugging

Key Benefit

These techniques enable AI to perform complex, multi-step workflows beyond simple code generation.

to a programmer interface for an agent.

Geek Box 13.3: Markdown-driven quality gates: when agents run the production checklist

A key insight from agentic software engineering is that quality criteria, which used to require human inspection, can now be expressed as structured markdown instructions that an AI agent executes autonomously. The agent reads the checklist, runs the specified checks, collects all findings, and presents them for human review—combining the thoroughness of automated scanning with the judgment-level assessments previously exclusive to human reviewers.

Example: a production checklist for book chapters. This book uses a markdown-based production checklist that an AI coding agent runs before each chapter is finalized. A condensed excerpt:

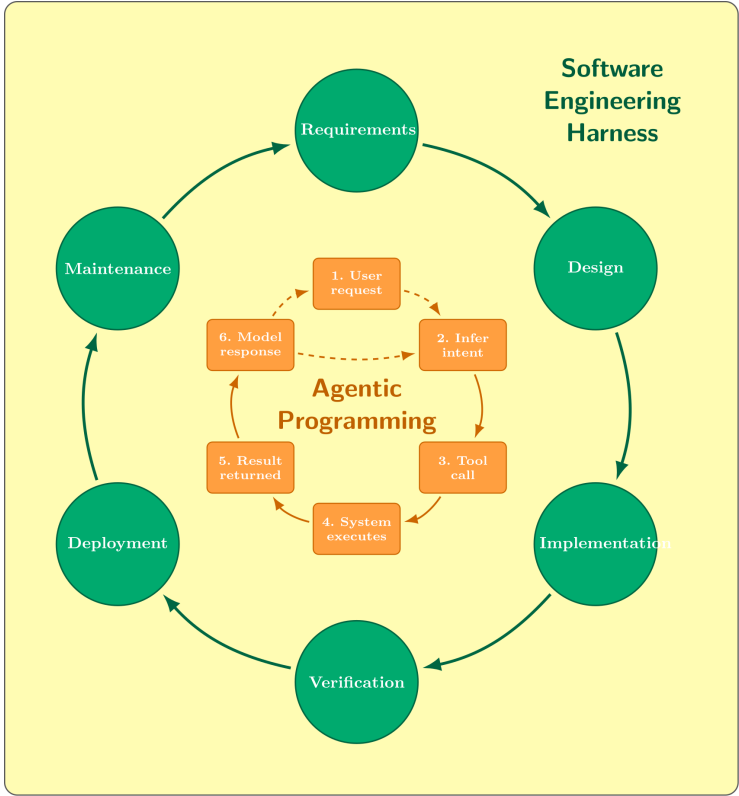
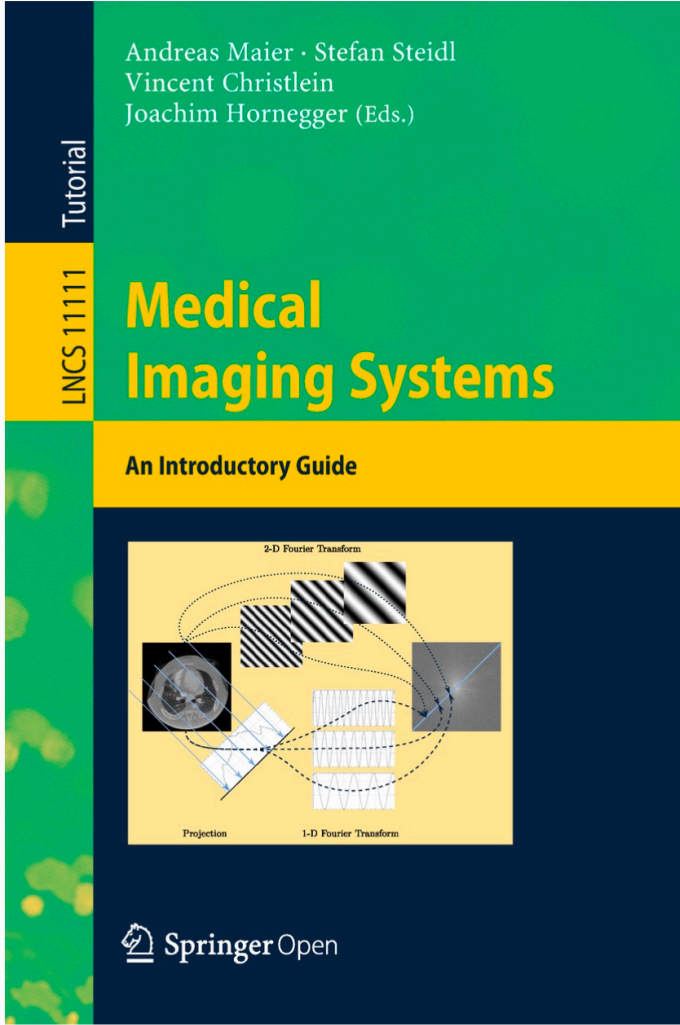
```
# Production Checklist for Chapter Review
The agent should: 1. Detect all problems first.
2. Present a PASS/FAIL table. 3. Ask the user
which to fix. 4. Apply only selected fixes.

## Checks to Run
### Cross-References
- All figures/tables/geek boxes referenced in text.
### Accessibility
- All figures have descriptive alt-text.
### Layout
- No "Float too large" or overfull hbox warnings.
### Acronyms
- No duplicate acronym expansions.
```

Several of these checks—alt-text quality, duplicate acronym detection, caption descriptiveness—require understanding the *meaning* of text, not just its syntax. An AI agent can evaluate whether alt-text actually describes what a figure shows or whether a manually spelled-out acronym conflicts with a `\gls{}` expansion nearby. These are assessments no simple script can perform reliably.

The detect-then-ask workflow. The checklist specifies a four-step process: detect all issues, present a PASS/FAIL summary, ask the user which fixes to apply, and then execute only approved changes. This separation of detection from correction mirrors the role-separation principle from Chapter 12: the agent that finds problems should not unilaterally fix them. The human decides which findings are genuine issues and which are acceptable.

This pattern generalizes beyond book production. Any domain with explicit quality criteria—code review checklists, regulatory compliance, accessibility audits, documentation reviews—can be encoded as markdown and executed by an agent. The quality gate becomes a reusable, version-controlled artifact rather than tacit knowledge in a reviewer's head.



What is coming next?

未来何去何从？

Everything is Easy Now 易如反掌

China and AI 中国与人工智能



Everything is Easy Now 易如反掌



Conference Shares 会议份额

Conference	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
CVPR	~30%	32%	35%	35%	39%	39.2%	~40%	~38%	~39%	~40%
ICCV/ECCV	~30%	30%	32%	35%	~37%	~40%	~40%	~38%	~40%	~40%
ICML	~10%	12%	15%	18%	~20%	~22%	~25%	~28%	~30%	~30%
NeurIPS	8%	10%	12%	15%	~15%	~18%	~20%	~25%	~28%	~30%
AAAI	15%	18%	22%	28%	~35%	50%	~50%	~45%	~48%	~50%

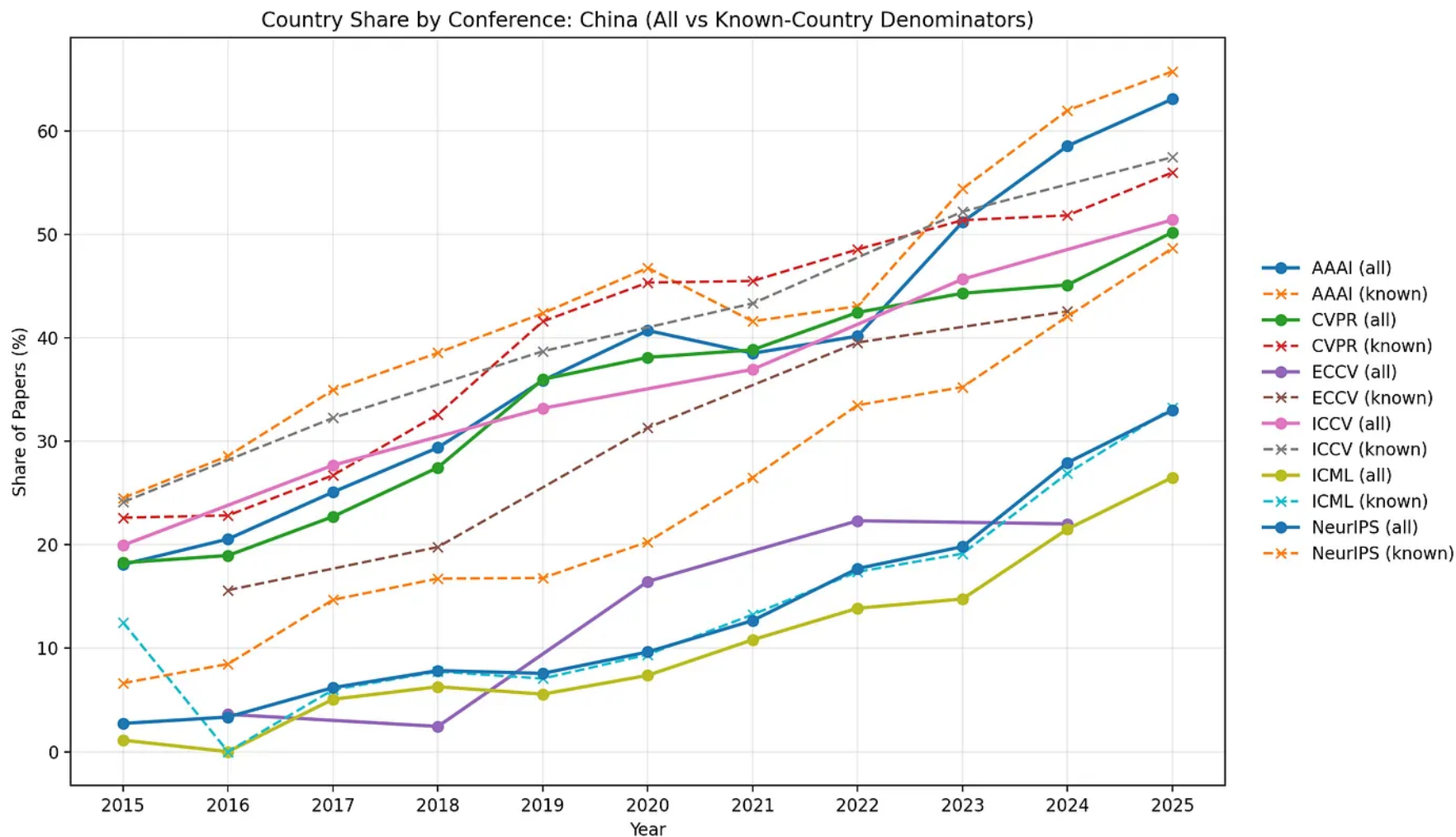
Everything is Easy Now 易如反掌

ConferenceStats Workflow ConferenceStats workflow



Everything is Easy Now 易如反掌

Conference Contributions from China 来自中国的会议贡献



- Continuously-running LLM agent that edits a CT reconstruction codebase, runs short GPU experiments, and keeps changes only when the metrics agree. Five agents run in parallel across five CT challenges and pool findings through a shared scratch pad.

Re-implemented solvers (pentathlon/demo_dl_reference):

- **Analytical:** PyroNN FBP (Syben 2019).
- **Classical iterative:** TV (iterative, supervised, hyper-search); Wu *et al.* 2015 (+ trainable port).
- **Variational / unrolled:** Hammernik 2017 variational network (+ CT port); Learned Primal-Dual (Adler & Öktem); ITNet v1–v3.
- **Self-supervised denoising:** dual-domain U-Net and trainable bilateral filters (Wagner 2023 / 2022, with supervised variants).
- **Per-scene neural fields:** NAF (neural attenuation field); R2Gaussian.
- **Foundation models:** RAM (Terris 2025).
- **Diffusion priors:** DDPM; diffusion-prior sampling; diffusion-recon with data consistency.
- **Transformer:** U-Swin.

github.com/akmaier/Agent4CT

- **Tight iteration budget.** 5 min per GPU experiment forces the agent to trade architecture against throughput.
- **Stage checks every 30 iterations.** 1 h validation on a $3\times$ larger subset catches overfit before it propagates.
- **Git-backed journal.** Every kept change is a commit; discards are `git reset` plus a row in the experiment TSV.
- **Anti-overfit rules** bootstrapped into every agent's `program.md` – so each new run starts on the others' shoulders, not from scratch.

- AAPM 2016 Low-Dose CT (Mayo)
- AAPM 2021 DL Sparse-View CT (128-view fan-beam)
- AAPM 2022 TrueCT
- AAPM 2022 DL-Spectral CT
- AAPM 2024 CT Metal Artifact Reduction

Score: *headroom recovered* $\in [0, 1]$; 1 matches the oracle.

Current best – Breast CT

Learned Primal-Dual
SSIM **0.9062**
at 1.49 M parameters

Foundation Models 基础模型

Bavarian Foundation AI Model 巴伐利亚基础大模型



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- D. Silver et al., *Mastering the game of Go with deep neural networks and tree search*, *Nature*, 529:484–489, 2016 (AlphaGo).
- K. Hammernik, T. Klatzer, E. Kobler, M. P. Recht, D. K. Sodickson, T. Pock and **F. Knoll**, *Learning a Variational Network for Reconstruction of Accelerated MRI Data*, *Magnetic Resonance in Medicine*, 79(6):3055–3071, 2018 (FastMRI / variational network).
- A. Maier, C. Syben, B. Stimpel, T. Wuerfl, M. Hoffmann, F. Schebesch, W. Fu, L. Mill, L. Kling, S. Christiansen, *Learning with Known Operators Reduces Maximum Error Bounds*, *Nature Machine Intelligence*, 1:373–380, 2019.
- M. Zaiss, J. R. Rajput, H. N. Dang, V. Golkov, D. Cremers, F. Knoll and **A. Maier**, *Exploring GPT-4 as MR sequence and reconstruction programming assistant: GPT4MR*, BVM Workshop, pp. 94–99, Springer, 2024.
- M. Zaiss, A. Aly, J. Endres, T. Dornstetter, S. Weinmüller and **A. Maier**, *Agentic MR sequence development: leveraging LLMs with MR skills for automatic physics-informed sequence development*, arXiv:2604.13282, 2026 (Agent4MR).
- Deutscher Zukunftspreis (low-field MRI): <https://www.deutscher-zukunftspreis.de/de/die-gewinner-des-deutschen-zukunftspreises-2023-spendeten-ein-mrt-geraet>
- Bavarian Foundation AI Model: <https://www.ai-bay.eu>
- Conference statistics tooling: <https://github.com/akmaier/ConferenceStats>
- Agent4CT – agentic re-implementation of deep-learning CT methods: <https://github.com/akmaier/Agent4CT>

Thank You & Congratulations! 致谢与祝贺



Congratulations to the **Department of Orthopedics, West China Hospital**
on **80 years** of healing, teaching and discovery.

谨祝华西医院骨科 八秩华诞！

杏林春暖，桃李芬芳；悬壶济世，再谱新篇。

Slides on GitHub 幻灯片见 GitHub



*From the early days of pattern recognition
to the future of medical AI –
thank you for joining this journey.*

温故而知新，止于至善。

github.com/akmaier/AIMed2026Slides/tree/wch-80-years-orthopedics